Section 1: Quantitative Aptitude (Aptitude & Speed)

1. A train moving at 72 km/h crosses a pole in 10 seconds. What is the length of the train?

(a) 150 m

(b) 200 m

(c) 250 m

(d) 300 m

2. If $x + y = 15$ and $xy = 56$, what is the value of $x^2 + y^2$?

(a) 113

(b) 121

(c) 115

(d) 105

3. A shopkeeper gives a 20% discount and still makes a 10% profit. If the cost price is 400, what is the marked price?

(a) 500

(b) 520

(c) 550

(d) 600

4. What is the average of the first 50 natural numbers?

(a) 25.0

(b) 25.5

(c) 26.0

(d) 24.5

5. The ratio of ages of A and B is 4:5. After 6 years, the ratio becomes 6:7. What is B's current age?

(a) 12

(b) 15

(c) 18

(d) 20

6. If 15% of x is equal to 20% of y, then $x:y$ is:

(a) 3:4

(b) 4:3

(c) 15:20

(d) 5:6

7. A sum of money doubles itself in 8 years at simple interest. What is the rate of interest?

(a) 10%

(b) 12.5%

(c) 15%

(d) 8%

8. If a work can be done by 10 men in 20 days, how many men are needed to finish it in 5 days?

(a) 30

(b) 40

(c) 50

(d) 25

9. What is the HCF of 24, 36, and 60?

(a) 6

(b) 12

(c) 18

(d) 24

10. A boat goes 10 km upstream and 10 km downstream in 3 hours. If the speed of the stream is 3 km/h, what is the speed of the boat in still water?

(a) 7 km/h

(b) 8 km/h

(c) 9 km/h

(d) 10 km/h

11. The price of sugar rises by 25%. By what percentage must a family reduce consumption to keep the expenditure same?

(a) 20%

(b) 25%

(c) 15%

(d) 30%

12. Find the missing number in the series: 2, 6, 12, 20, 30, ?

(a) 40

(b) 42

(c) 44

(d) 46

13. A circle has an area of 154 cm^2. What is its circumference? ($\pi = 22/7$)

(a) 22 cm

(b) 44 cm

(c) 66 cm

(d) 88 cm

14. If $2^x = 32$, what is the value of x^2?

(a) 10

(b) 16

(c) 25

(d) 36

15. A man buys an item for 120 and sells it for 150. What is his profit percentage?

(a) 20%

(b) 25%

(c) 30%

(d) 15%-----Section 2: Data Structures & Algorithms (Easy - Med)

16. Which data structure works on the LIFO (Last In First Out) principle?

(a) Queue

(b) Linked List

(c) Stack

(d) Tree

17. What is the time complexity of searching an element in a balanced Binary Search Tree (BST)?

(a) $O(1)$

(b) $O(n)$

(c) $O(\log n)$

(d) $O(n \log n)$

18. Which of the following sorting algorithms is NOT stable by default?

(a) Merge Sort

(b) Insertion Sort

(c) Quick Sort

(d) Bubble Sort

19. To implement a Min-Heap using an array, if a parent is at index i, what is the index of its left child?

(a) $i + 1$

(b) $2i$

(c) $2i + 1$

(d) $2i + 2$

20. What is the space complexity of a recursive Depth First Search (DFS) on a graph with V vertices?

(a) $O(1)$

(b) $O(V)$

(c) $O(E)$

(d) $O(V + E)$

21. In a Singly Linked List, what is the time complexity to delete a node at the end (given only the head pointer)?

(a) $O(1)$

(b) $O(\log n)$

(c) $O(n)$

(d) $O(n^2)$

22. Which data structure is best suited for implementing a Breadth First Search (BFS)?

(a) Stack

(b) Queue

(c) Priority Queue

(d) Hash Map

23. What is the worst-case time complexity of Quick Sort?

(a) $O(n \log n)$

(b) $O(n^2)$

(c) $O(n)$

(d) $O(\log n)$

24. A hash table with chaining avoids collisions by:

(a) Using a second hash function

(b) Finding the next empty slot

(c) Creating a linked list at the index

(d) Increasing the table size immediately

25. Which traversal of a Binary Search Tree (BST) produces a sorted sequence of elements?

(a) Pre-order

(b) Post-order

(c) In-order

(d) Level-order

26. What is the maximum number of nodes in a binary tree of height h (where the root is at height 0)?

(a) 2^h

(b) $2^h - 1$

(c) $2^{h+1} - 1$

(d) 2^{h-1}

27. In a Priority Queue, which data structure is most commonly used for efficient $O(\log n)$ insertion and deletion?

(a) Sorted Array

(b) Linked List

(c) Heap

(d) Binary Tree

28. What is the main advantage of a Circular Queue over a Linear Queue?

(a) Faster access time

(b) Better memory utilization

(c) Easier to implement

(d) Can store more elements than capacity

29. Which algorithm is used to find the shortest path from a single source to all other nodes in a weighted graph (no negative weights)?

(a) Prim's Algorithm

(b) Kruskal's Algorithm

(c) Dijkstra's Algorithm

(d) Floyd-Warshall Algorithm

30. What does the "Dynamic" in Dynamic Programming refer to?

(a) The memory is allocated dynamically

(b) Sub-problems are solved and their results are re-used

(c) The program changes its code at runtime

(d) It only works on dynamic data structures